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Research article

System and method to diagnose conjunctivitis in the eye of a user ☆

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A R T I C L E I N F O

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A B S T R A C T

This Proposed work explores how machine learning can be used to diagnose conjunctivitis, a common eye ailment. The main goal of the study is to capture eye images using camera-based systems, perform image pre-processing, and employ image segmentation techniques, particularly the UNet++ and U-net models. Additionally, the study involves extracting features from the relevant areas within the segmented images and using Convolutional Neural Networks for classification. All this is carried out using TensorFlow, a well-known machine-learning platform. The research involves thorough training and assessment of both the UNet and U-net++ segmentation models. A comprehensive analysis is conducted, focusing on their accuracy and performance. The study goes further to evaluate these models using both the UBIRIS dataset and a custom dataset created for this specific research. The experimental results emphasize a substantial improvement in the quality of segmentation achieved by the U-net++ model, the model achieved an overall accuracy of 97.07. Furthermore, the UNet++ architecture displays better accuracy in comparison to the traditional U-net model. These outcomes highlight the potential of U-net++ as a valuable advancement in the field of machine learning-based conjunctivitis diagnosis.

1. Introduction

Eyes Conjunctivitis or pink eye, is a highly common eye problem that impacts people of all age groups and can be caused by ample reasons. The conjunctiva is the clear membrane that lines the eyelid and covers the white part of the eye. When this membrane becomes inflamed, it results in the symptoms of conjunctivitis. These symptoms include redness, itching, burning, and discharge from the eye. The discharge can be clear, white, yellow, or green; in some cases, the eye may become swollen and crusted over. Three variants of conjunctivitis are viral, bacterial, and allergic. Viral conjunctivitis is caused by a viral infection and is highly contagious. This type of conjunctivitis is usually accompanied by other symptoms such as a runny nose, sore throat, and fever. Bacterial conjunctivitis is caused because of a bacterial infection. It is characterized by a thicker and more profuse discharge from the eye and often requires antibiotic treatment. Allergic conjunctivitis is one of the eye conjunctivitis that occurs due to an allergic reaction to a constituent such as pollen, dust, or makeup and is not contagious. It is characterized by itching, redness, and tearing. Conjunctivitis can also be caused by irritants such as smoke, chlorine, or other chemicals. In these cases, the condition is referred to as irritant conjunctivitis. For viral and bacterial

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conjunctivitis, antiviral and antibiotic medications may be prescribed. For allergic conjunctivitis, antihistamines and artificial tears may be used to alleviate symptoms. In some cases, simple measures such as applying a warm compress to the affected eye can also help to relieve symptoms. It is important to note that conjunctivitis can be highly contagious, so it is important to maintain good hygiene such as cleaning hands frequently and avoiding touching the eyes to prevent the spread of the infection. If symptoms persist or if vision is affected, it is important to see an eye doctor for proper diagnosis and treatment. In most cases, conjunctivitis is not serious and resolves on its own within a few days to a week. However, if left untreated, it can lead to more serious complications such as corneal ulcers or scarring. Therefore, it is important to seek prompt medical attention if you suspect that you have conjunctivitis. With proper diagnosis and treatment, most cases of conjunctivitis can be effectively managed, and full recovery can be achieved.

This paper proposes a way to deal with the spread of conjunctivitis and real-time identification of contagious eye diseases using ML. Machine Learning gives various approaches to identifying a distant object using image processing to simplify the complex task and can be used to identify contagious eye diseases. The key idea is to capture the image of the eye using a camera. A person needs to gaze into the camera lens. Then pictures of the eye will be captured for further processing. The snap will be used to extract features from the eye image.

2. Related work

Conjunctivitis is one of the most common and harmful contagious eye diseases. Face and eye detection or recognition are among the topmost scrutinized fields of computer science. Many papers have proposed various approaches to avoid the spread of conjunctivitis. Most of the papers discussed studying the sclera of the eye. Segmentation of the iris from the sclera and deep studying of blood vessels. Grab-cut method and edge-cutting algorithms are used for the segmentation of the iris from the sclera. The most popular open-source computer vision library today is OpenCV. It includes various pre-trained and already existing classifiers for face recognition and eye detection.

2.1. Literature review

1. In paper [1] randomly distributed blood veins inside the sclera part of the eye are studied. The proposed work comprises new methodologies for eye detection, sclera segmentation, vessel enhancement, extraction, and binarization. UBIRIS.v1 and UTIRIS datasets are used for achieving segmentation accuracy and computational complexity. Two strategies were employed for automatic scleral segmentation: scleral pixel threshold and scleral shape contouring. The focus of these papers is a technique called Scleral Shape Contour, which is used to remove noise and segment the scleral region. The sclera is automatically identified without any supervision from a person. Utilizing seed-based contour initialization and Active Contours without Edges (ACWE) for the segmentation of the sclera by delineating the eye's boundaries. The limitation of these contour techniques lies in their dependency on an edge function that fluctuates with variations in image gradient.
2. In paper [2] various Sclera feature extraction techniques have been discussed and a comparison between techniques has been made based on their accuracies.

1. Hu's invariant moments, Minutiae

It segments the sclera from the eye image using CLAHE (Contrast Limited Adaptive Histogram Equalization). The Hu invariants are scale, rotation, and translation invariants that is they remain unchanged under these types of transformations.

2. Discrete Cosine Transform

A DCT represents an image as a sum of cosine functions of different frequencies, allowing the most important information of an image to be represented with fewer coefficients. Enhance conjunctival vascular patterns to extract features from visible vascular patterns.

3. Affine Transformation

It is used to change the position, orientation, and shape of an object in 2D or 3D space for image wrapping registration and shape matching. The feature vector (m) is transformed using an equation. (1) given below:

$$m_7 f(xe, y, t) + m_8 = f((m_1 x + m_5 + m_2 y), (m_3 x + m_4 y + m_6), (t - 1)) \tag{1}$$

4. Line segment descriptor

It is used for image matching, object recognition, and scene understanding. It uses a canny-based edge detection method and Gabor filter-based enhancement is applied to enhance the sclera. It converts sclera-segmented images into a binary image and results in segmented sclera images in grayscale.

5. Harris corner and edge detection

Interaction points of vessels are calculated, and some reis transformation is performed to extract information. To calculate the interaction point, the first derivative is calculated, and then the noise is removed from the image. A corner formula is applied and finally find local maxima from the corner response.

6. DWT Co-efficient

The image is represented using a discrete wavelet transform coefficient. For the same, the segmented 2D image is converted to 1D and the coefficient is to be calculated using equation. (2) given below:

$$Energy(I) = \sum_{r=1}^{N-1} \sum_{c=0}^{N-1} (I_{r,c})^2 \tag{2}$$

4. This study [3] presents an original methodology for portioning the sclera, which depends on an improved rendition of the U-Net model, alluded to as Sclera-SegNet. The exploration includes a point-by-point assessment of the U-Net design's construction and

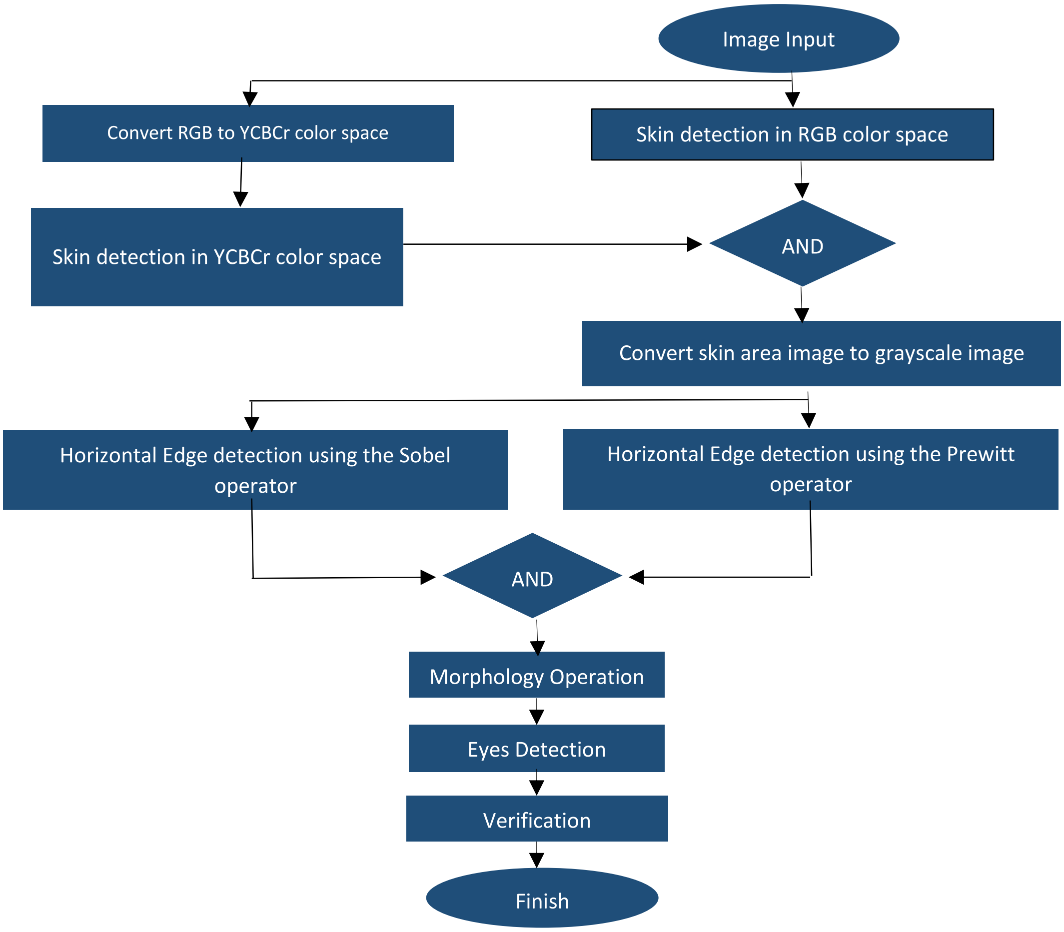

Fig. 1. Proposed method architecture.

6. In paper [9] face and eye detection approaches are used to separate face and eye so that analysis of the image of the eye can be performed. The paper uses a cascade multitask convolutional network. They were able to achieve a 98 % percent accuracy when used with the available datasets. The method proposed here is MTCNN and is powerful enough to identify the image of the eye and segment the eye from the image of a face.
7. In the paper [10] an approach to Adenoviral conjunctivitis is discussed. Image processing has contributed to the cure of diseases with powerful features of improving the efficiency of an image. Adenoviral conjunctivitis or pink eye is redness and inflammation of membranes inside the sclera part and an increase in vascularization on the sclera. The approach used is to use the Grab-Cut segmentation method for segmenting an image of the eye and then Machine learning classifiers are used to see the presence of conjunctivitis.
8. In paper [11] sclera is studied from different angles. So that each corner of the eye sclera can be studied and analyzed. The focus is on studying the frontal and side sclera. In Fig. 2 there are two images of eyes. Image (a) shows side looking image of the eyes and (b) shows a frontal-looking image of the eyes. A comparison is made between (a) and (b) to find a clear picture of both by applying various quality measures.

recommends the coordination of a consideration module inside the focal bottleneck section arranged between the contracting and far-reaching pathways of U-Net. This expansion is pointed toward upgrading the organization's ability to learn unmistakable highlights. The examination incorporates a correlation of different consideration modules, eventually discovering that the channel-wise consideration module is the best in upgrading the division organization's presentation. Furthermore, the review evaluates the effect of an information expansion process in improving the division organization's speculation capacities.

5. In paper [4] eye recognition using edge detection and Euclidian distance. The edge detection is undertaken using Sobel and Prewitt methods. Skin is identified using a skin-revealing procedure that uses formulas for two different color spaces. Very small edges are removed using morphology operations. Euclidean distance method. The outcomes of the paper in detecting the eye and separating the eye from the face gave an accuracy of 93 percent. Fig. 1 shows the Architecture of the proposed method.

9. In the paper [5], a new depth-sensing model has been proposed that considers affine transformations to compare two vascular structures. Scleral region segmentation, scleral vessel pattern extraction, gazing direction identification, and comparison among the two vessel patterns for coordinating and identification. The anticipated segmentation model, called DSeg, lowers the complexity by constructing a base of knowledge for scleral and not sclera colours and leverages the U-Net deep learning model.
10. In the paper [6], the process of scanning the eye region begins by dividing it into several blocks with different positions and ranges. An LBPH is obtained from each cell block to generate a local texture descriptor. A reference template is then created for each block. This is the optimal histogram that maximizes the reparability between blocks and her LBPH in different clusters. The bitwise distance between the local LBPH and the corresponding reference template is then calculated for every block to structure the set of features for classification. Finally, we use a cascaded Ada-Boost learning procedure to pick the furthermost discriminating features from the features set speeding up the classification process. Below are the results of the work performed in the paper.
11. In paper [7], the purpose of this landmark article is to explore the effectiveness of using convolutional neural networks (CNNs) for scleral detection by building a neural model that can be trained for this purpose. This study uses the SSRBC 2015 dataset. This dataset contains 734 images taken from distinct positions from 30 unique classes. The intended method demonstrates the ability of neural learning in developing scleral detection systems. Fig. 3 shows the steps involved in the process of sclera segmentation.

12. In paper [8], a blood vessel extraction algorithm for sclera-conjunctival images can be implemented for syndrome discrimination by one-eye observation. Sclera-conjunctival regions are isolated using optimal threshold segmentation and mathematical manipulation. In addition, scanning and edge detection techniques are applied to identify vessel edges. Using these methods, the system can extract edge feature parameters that can be used for vessel reconstruction. Experimental results demonstrate that the algorithm can extract vascular information quickly and accurately. The algorithm is implemented using MATLAB and includes the conversion of true-colours images of sclera-conjunctival regions to binary images by grayscale and binary processing, automatic selection of optimal thresholds, and removal of spurious vessels, and vessels of the scleral-conjunctival region. Fig. 4 shows the framework of the process.

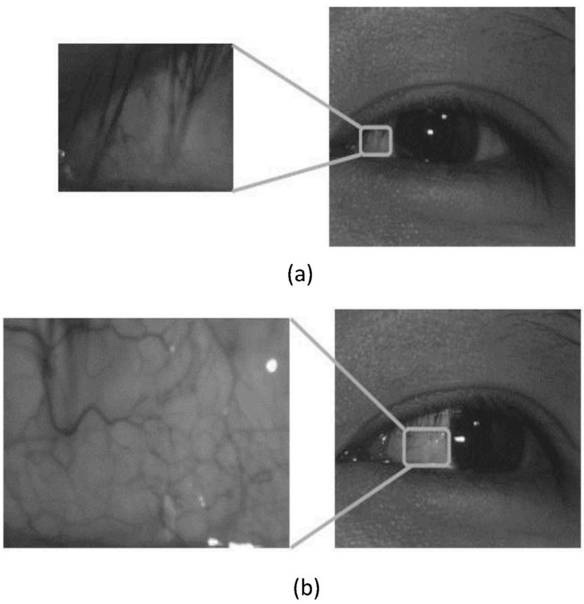

Fig. 2. Sclera image from multiple angles. (a) Side looking image (b) frontal-looking image.

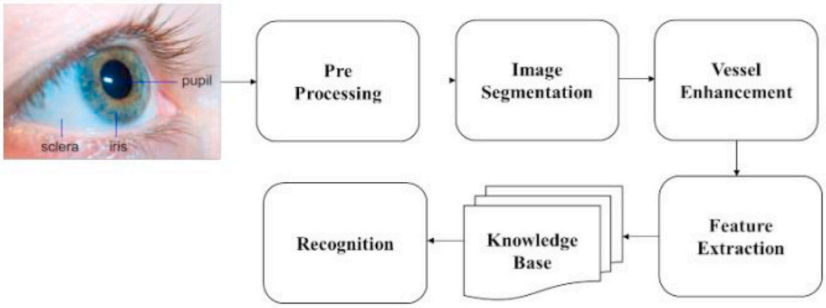

Fig. 3. Sclera segmentation and recognition process.

2.2. Results from related work

1. The approach used in Ref. [2] uses existing vision and ML technology. CNNs have been presented in papers that segment the sclera and Iris from previous eye images. Features are obtained from segmented regions such as iris diameter or radius proportion, scleral mean reddening index (MRL), and reddened area ratio (RAP), and some features are extracted using scleral contours. At least six eye features are taken out of eye images and a binary classifier is used to estimate IOP risk for training and image testing purposes. Offering a new foundation to help two ophthalmologists and extreme IOP competitors with initial IOP is inappropriate because it is an extremely active approach for fear of total or partial fantasy failure due to IOP or glaucoma. We introduce a new concept-specific structure, structure, and layout for triggering IOP masks using frontal eye representations. This work shows that there is proof of computational friendliness between IOP classification and eye frontal appearance. The new scleral contours get work here that wasn't there before, introduced in the frontal eye imaging literature for IOPOP class persistence. We present a new application for a Fully Convolutional Neural Network (FCNN) in sclera and iris recognition and division. Improving the classification by applying a decision tree (DT) classifier to the field obtained from the updated scleral segment.

Unfortunate lighting, a bothered sclera, a shut-eye, serious obscuring, a non-centered, uncropped eye region, and different issues add to the low quality of certain pictures. Both single-meeting and multi-meeting settings have been utilized to finish the assessment cycle. They involved three pictures for preparing and two for testing for every client in the single-meeting setting, while in the multisession situation, they involved pictures from meeting 1 for testing and those from meeting 2 for preparing.

The correct sclera validation (CSV) [1] rate is calculated by subjectively comparing the correct validation images to the eye image decision and is utilized to evaluate the effectiveness of the proposed sclera validation. CSV is calculated by equation 3

$$CSV = ((Number\ of\ CAS + Nymber\ if\ CRS) \div Total\ number\ of\ images) \times 100\% \tag{3}$$

Table 1 shows the outcome of session 1 and session 2.

The accuracy of the proposed methodologies was able to achieve 98.65 % for the first session and 95.3 % for the second session which is higher than any other active contours methods. In comparison with other methods, the proposed method also reduces processing time am 0.003 for session 1 and 0.010 for session 2, respectively. Table 2 shows the time complexity achieved by the proposed system.

On the IRIS database, various active contour model sclera segmentation methods are compared in terms of accuracy and complexity. The Geodesic Active Contour [1] Method segmented images of the sclera with an accuracy of 66.33 % and a processing time of 3.145s. Ballons Active Contour Method segmented a sclera image with an accuracy of 71.01 % and a processing time of 3.708s. GVF Active Contour Method segmented a sclera image with an accuracy of 82.44 % and a processing time of 4.779s. The approach used in the proposed work outperforms other contour methods. Proposed Active Contours without Edge segmented clear image with an accuracy of 90.82 % and processing time of 0.004s.

2. Discussed various Sclera feature extraction methods namely Hu's invariant moments, Discrete Cosine Transform, Affine Transformation, Line segment descriptor, and Harris corner and detection. The feature extraction was performed in two different modes, one is a direct method and the other one is an Error equal rate method. Two methods Hu invariants and affine transform-based feature extraction method were tested by direct method. The database used for Hu's invariant and affine transform is a house database, resulting in an Error Equal rate of 100 % for HU's and 25 % for Affine transform. Table 3 shows the methods of feature extraction with the corresponding databases, the number of samples, and the percentage error, and Fig. 5 bar graph represents the Error effective rates of different methods.

   Line segment descriptor, and Harris corner and detection method were tested using Error Equal Rate methods and UBIRIS database.

3. The proposed strategy includes the use of the first U-Net model as a standard. Consequently, four improved U-Net variations, each furnished with particular consideration modules, are thought about against the standard. Aside from changes in network structure,

any remaining parts of the organization stay steady. Preparing and testing are directed at datasets, including UBIRIS.v2, MICHE-I, and three subsets of MICHE-I, with assessments given PR-bend and F-measure measurements, as portrayed in Fig. 6. The outcomes uncover that the superior U-Nets highlighting different consideration modules fundamentally outflank the benchmark model on (b) MICHE-I, (c)MICHE-GS4, (d)MICHE-IP5, (e)MICHE-GT2, and (f)MASD.v1. On account of UBIRIS.v2, there is negligible divergence in PR-bends and F-measure values between the benchmark model and its improved adaptations. Curiously, the better U-Nets, expanded with different consideration modules, display reliable PR-bends and F-measure values for UBIRIS.v2, MICHE-I, and their subsets. Notwithstanding, these models yield altogether unique division results for MASD.v1. Outstandingly, the U-Net improved with CBAM accomplishes the most elevated division execution, achieving a F-measure worth of 83.77 %, while CAM, BAM, and SAM follow intimately with F-measure upsides of 82.55 %, 80.67 %, and 77.78 %, individually. Upon closer assessment, it becomes apparent that channel-wise consideration holds more prominent importance than spatial-astute consideration in guaranteeing division exactness, which is the reason numerous other division networks have likewise taken on channel consideration modules to upgrade network execution.

5. The MTCNN model is utilized in the main examination. The Haar Fountain Calculation is utilized for the dataset in the second examination too. An examination of the suggested face-and-eye acknowledgment models utilizing MTCNN and Haar Outpouring is displayed to make sense of the proposed model after the dataset is executed as a test information assortment. The precision accomplished with MTCNN is 98 %, while that with Haar Fountain is 68.16 %.

Table 5 shows the results of various algorithms that are used for eye detection and the proposed algorithms are compared. The Viola-Jones algorithm is used on the MIT dataset with a detection rate of 61.8 %. The Hough Transform algorithm is used on the Casia Dataset with a detection rate of 83 %. Multi-task Cascaded Convolutional Networks (MTCNN) used on the NICE-II and MICHE dataset gave a detection rate of 92 % and the Haar Cascade Algorithm that used a set of sample images collected from different sources and gave a Detection rate of 94 %.

6. The training set consisted of 12 Ad-Cs and 18 healthy eye images. Calculated Redness, Vascularization, and GLCM for the feature set. Images of healthy and Ad-Cs eyes from the training set. The Ad-Cs have significantly higher values for vascularization and redness than healthy eyes. Ad-Cs, on the other hand, had a lower standard deviation for GLCM than Healthy ones as shown in Tables 6 and 7.

Fig. 7 shows results including GLCM values and Pictures of sound and Promotion Cs eyes from the preparation set. The Promotion Cs have fundamentally higher qualities for vascularization and redness than solid eyes. Promotion Cs, then again, had a lower standard deviation for GLCM than Sound ones. The Promotion Cs and Solid ones' GLCM values differ essentially across the preparation set. With 6-overlap cross-validation and the ML strategies for Bayes and Arbitrary Tree, just 86 % of expectations are precise. Following the expulsion of the GLCM highlight from the preparation set and the utilization of the stochastic slope descent [11] (SGD) procedure with six-fold cross-approval, 29 examples were accurately characterized and one example was erroneous, yielding an exactness of 96.7 %, as displayed in Table 7. Notwithstanding, using just RGB the assumption precision rate was 90 %. The proposed approach had the option to pick the seed for the GrabCut with a more noteworthy precision and became free of goal considering the fact size of as far as possible

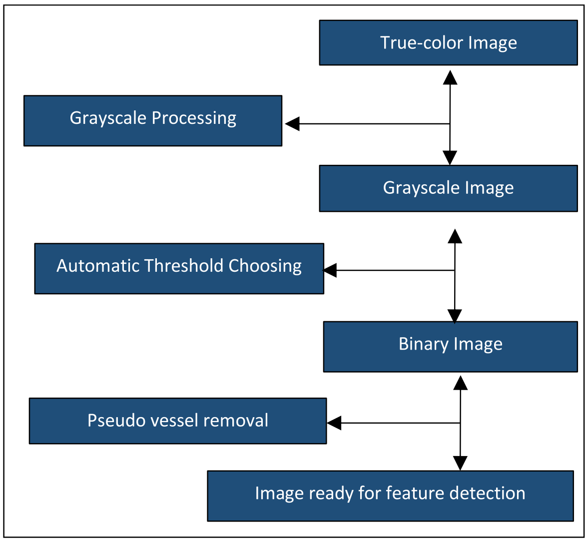

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Fig. 4. Process framework.

13. [12] Current cutting-edge image segmentation models, such as U-Net and FCN, employ encoder-decoder architectures with skip connections. These connections blend high-level semantic information from the decoder with fine-grained details from the encoder, proving effective in generating precise segmentation masks, even in complex contexts. This approach is crucial for instance-level segmentation models like Mask RCNN. While these models excel in natural image segmentation, they may fall short of meeting the stringent demands of medical image segmentation. Detecting lesions or abnormalities in medical images requires a higher level of accuracy. Even minor segmentation errors can negatively impact clinical diagnoses and user experience. To address this, we introduce U-Net++, a novel architecture with nested and dense skip connections. U-Net++ gradually enriches high-resolution encoder feature maps before fusing them with semantically rich decoder feature maps, making it more effective in capturing fine details. In contrast, traditional skip connections in U-Net directly pass feature maps, resulting in the fusion of semantically dissimilar maps. Our experiments demonstrate that U-Net++ significantly outperforms U-Net and wide U-Net, making it a promising solution for accurate medical image segmentation.
14. This study [13] intends to sort eye illnesses, explicitly typical, waterfall, and glaucoma, using a Convolutional Brain Organization (CNN) with the Efficient-Net design, explicitly EfficientNet-B0. The dataset, obtained from Kaggle, includes 300 pictures, which are expanded to produce 3600 pictures, conveyed across "ordinary" (1200 pictures), "waterfall" (1200 pictures), and "glaucoma" (1200 pictures). Four particular datasets are made from this information: the first dataset, an expansion dataset, an expansion dataset pre-processed in grayscale, and an increase dataset pre-processed with thresholding. The ideal exhibition is accomplished utilizing the Adam enhancer, a learning pace of 0.00001, and a cluster size of 32, with preparing led north of 20 ages. The best dataset is distinguished as the pre-processed grayscale expansion dataset, yielding vital outcomes with 79.22 % exactness, 80.3 % accuracy, 79.22 % review, and 78.87 % F1-Score.
15. The point is to make a methodology that utilizes profound convolutional brain networks which will run for the two eyes in lined up for visual component extractions. Dataset pictures for preparation and approval were acquired from the eye figment dataset [14]. Testing is finished by contributing constant recordings. Different plan and execution imperatives incorporate the inaccessibility of the predefined informational index, the excellent camera required, and preparing information models waiting to be refreshed like clockwork in collab. The presumptions and conditions are cameras will catch clear pictures from which we can extricate eye highlights, then on involving GPU in collab handling power is there contrasted with a computer chip, adequate lighting conditions are accessible, likewise adequate dataset is accessible for preparing.
16. The proposed [15] work utilizes Res-UNet given the design of the U2Net organization and utilizes the Information Improvement Tool compartment in light of little datasets, in which the Res-U2Net organization (given PyTorch) depends on a lot of information expansion to utilize the accessible explanation tests all the more effectively, and replaces the skip-association module with leftover delicate in light of the first U2Net The association module, which, as well as lessening the semantic distinction between low-level and significant level elements, convolutional include extraction can likewise be prepared to decrease commotion passed from low-level elements to undeniable level highlights. At long last, the component blocks after sound decrease are melded with the significant level elements. At long last, the number of organization boundaries and induction time are utilized as assessment pointers to assess the model. Using Miou, Precision, Recall, F1-Score, and FLOPS, various eye data segmentation frames were compared simultaneously. Due to the excessive number of parameters, experimenting with a small-scale U2Net combined with a Res module with a parameter volume of 4.63 MB, which is similar to U2Net in related indicators, verifies the effectiveness of our structure, which achieves the best segmentation effect in all comparison networks and lays a foundation for the application of subsequent visual apparatus recognition symptoms. Miou reaches 97.8 %, S-measure reaches 97.7 %, and F1-Score reaches 99.0
17. A clever pipeline for paleness assessment comprising of three principal commitments: a sclera division calculation applied to approach taken computerized photographs of the eye, a vessel extraction calculation, and a classifier to foresee the sickly status of an individual versus typical controls. This study depended on the public dataset Eyes-resist sickliness, which contains 218 eye pictures taken with an exceptional gadget that eliminates any surrounding light impact. Extremely intriguing outcomes have been accomplished for the sclera division task with great accuracy (88.53), review (82.53), and F1 (84.10). The variety highlights and haemoglobin esteem have all the earmarks of being very much related permitting us to get a F2 score in the iron deficiency location undertaking of 86.4 % utilizing variety highlights from the entire sclera and of 83.8 % utilizing just vessel variety highlights. Based on the pallor of the tissues examined in this study, several classification algorithms have been put through their paces to estimate the anaemic condition. The best-performing SVM is an assist vector with machining with a polynomial digit. The results were gained by separating the hemoglobin values into 2 classes according to the WHO edge for wiped-out/non-iron inadequate conditions.
18. The paper [16] gives an outline of the 2020 Sclera Division Benchmarking Rivalry (SSBC), the seventh contest in a progression of gathering benchmarking contests fixated on the Sclera Division issue. As opposed to past versions, the goal of SSBC 2020 was to assess the viability of sclera-division models on cell phone pictures. The resistance was used as a phase to assess the responsiveness of existing models to I) contrasts in phones used for picture catch and ii) changes in the encompassing obtainment conditions. 26 investigation packs enlisted for SSBC 2020, out of which 13 participated in the last round and introduced an amount of 16 division scoring models. These included a single method based on standard picture handling procedures and a wide range of deep learning arrangements. The experiments made use of three recent datasets. A large portion of the division models accomplished somewhat steady execution across pictures caught with various cell phones (with slight contrasts across gadgets), yet battled most with bad quality pictures caught in testing encompassing circumstances, i.e., in an indoor climate and with unfortunate lighting.

Table 1
CSV rate on UBIRIS.v1.

CSV	Season 1	Season 2
	95.38 %	89.45 %

Table 2
Time Complexity of the proposed system.

Sclera Recognition Steps	Complexity
Iris Segmentation	1.97s
Sclera Validation	0.24s
Sclera Segmentation	0.007s
Feature Extraction	0.29s
User Template Registration	0.46s

Table 3
Different feature extraction methods and their performance.

Feature Extraction method	Database	No. of Samples	% or EER
Hu's	In house	6/12	100 %
CDF	In house	50/300	4.3
Affine	In house	50/100	25 %
Line	UBIRIS.v1	241/1205	3.05
Harris	UBIRIS.v1	241/1350	2.19
DWT co-	UBIRIS.v1	241/1350	0.021429

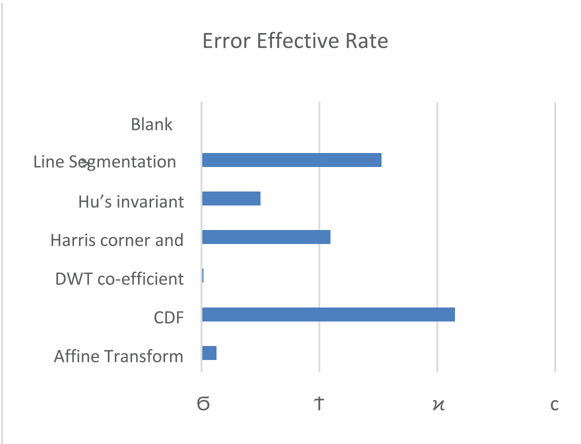

Fig. 5. Different feature extracted Method's EER

the region of the eye locale. The division step is more grounded when GrabCut is utilized with Redness and Vascularization since lightning and skin variety never again present difficulties. Subsequently, the proposed framework is economical, easy to set up, and powerful.

Next, we will examine the models that have been wrongly categorized and discover that the adaptive GrabCut algorithm [11] is unable to isolate the sclera region's boundary, resulting in greater variation. It concluded that the grab-cut algorithm and Bezier-type curves could more precisely isolate the sclera regions in eye images.

7. Multi-point sclera acknowledgment is completed involving left-looking and front-facing-looking pictures in this review. Purple (420 nm), Blue (470 nm), Green (525 nm), Yellow (590 nm), Orange (610 nm), Red (630 nm), Dull Red (660 nm), and Infra-Red (820 nm) are the eight frequencies that light up the IUPUI multi-recurrence database. At every frequency, there is a sum of 352 pictures of 44 subjects with five eye tones — blue, dull brown, light brown, green, and hazel. Every human subject's picture and recording were acquired on two separate events from the two eyes at six unmistakable points — left, right, up-left, up-right, front, and up. Between every information assortment, seven days pass.

The blue and red lines in underneath figure, which address two distinct vessel formats, associate the coordinating matches with green lines. The shade of the green lines demonstrates the strength of the match; while a coordinate with a blunter tone is more vulnerable, one with a more splendid tone is more grounded or more confident. The accompanying Fig. 8 shows how well two front-facing looking and two remaining-looking sclera pictures of a similar eye coordinated.

The bad match between two frontal-looking sclera pictures of the same eye and the good match between two left-looking pictures of the sclera are depicted in Fig. 9.

For Sclera acknowledgment Examination of front-facing-looking and left-looking sclera acknowledgment precision utilizing four

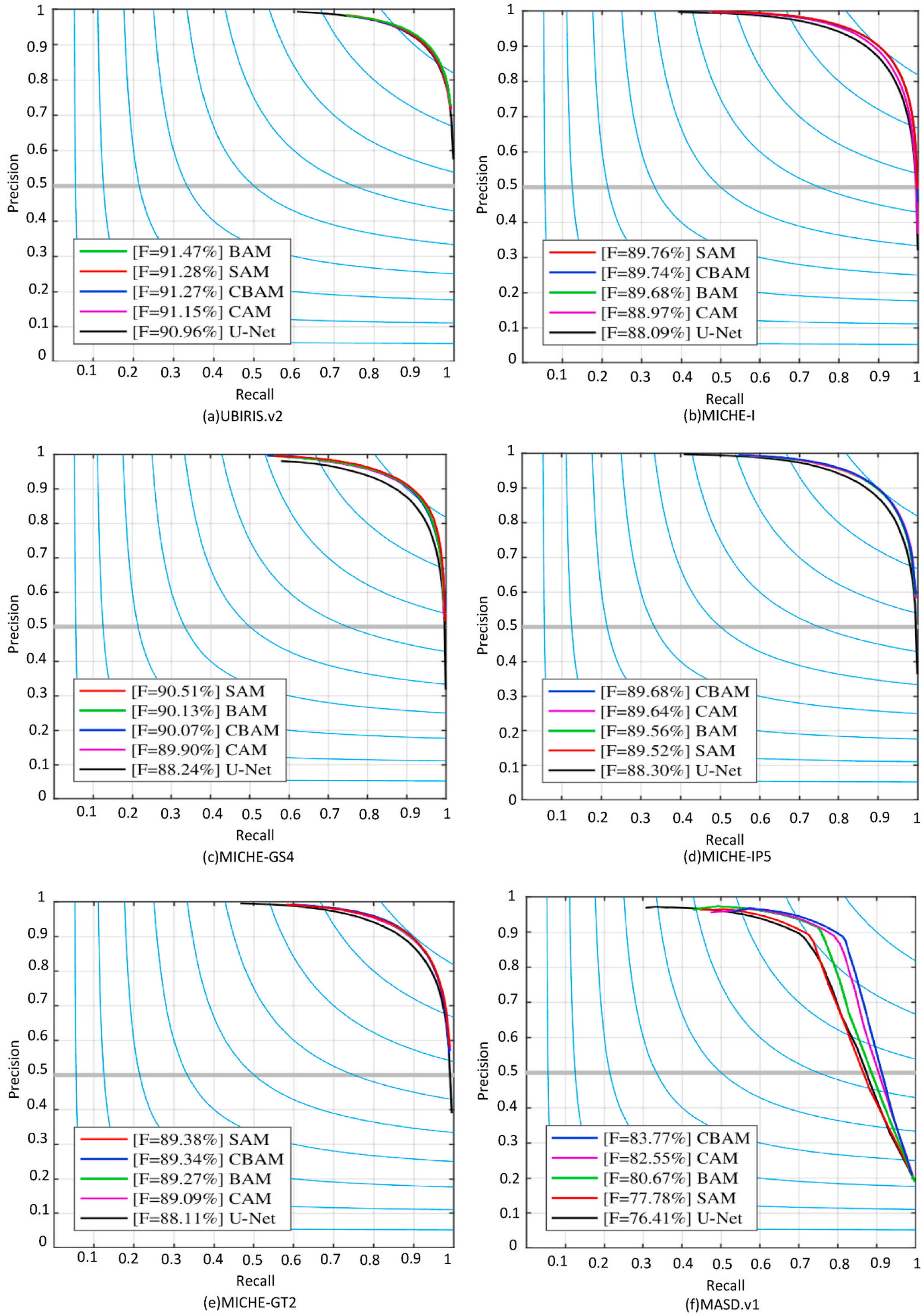

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Fig. 6. On six datasets, the average precision-recall curves produced by U-Net and improved U-Nets with various attention modules. (a) UBIRIS.v2 developed for iris recognition in less constrained conditions. (b) MICHE-I was developed for mobile iris recognition and its subsets (c), (d), (e). (f) SBVPI is a publicly available database.

4. The proposed method uses the PICS database for image datasets with different sizes and lighting conditions. The database includes images of both men and women from different races and ages. The results from the proposed methods outperform the methods based on edge density. The methods based on edge detection give an accuracy of 80 % and the proposed method gives an accuracy of 93 % as shown in Table 4.

Table 4
Experimental results on PICS image databases.

Method	Accuracy
A method based on edge density	80 %
The proposed method	93 %

Table 5
Examination of discovery pace of eyes among different techniques for various datasets.

Algorithm	Detection rate	Dataset
Viola Jones	61.81 %	MIT
Hough Transform	83 %	Casia Database
MTCNN	92 %	NICE-II and MICHE database
Haar Cascade	94 %	Set of 10000 test images

Table 6
Chosen feature of healthy and Ad-Cs Eye for the representative set.

Sample	Redness	Vascularization	Std for GLCM
Healthy	0.0250	0.07084	48.3567
Ad-Cs	0.2409	0.14045	39.8786

Table 7
Confusion Matrix for SGD with 6-fold cross-validation.

Sample	Healthy	Ad-Cs
Healthy	17	1
Ad-Cs	0	12

combination techniques — the straightforward normal, max-score, min-score, and quality-based normal combination strategy [5]. To look at the acknowledgment exactness of the different methodologies, this information base has been portioned physically and consequently.

In manual division, the general precision of value-based normal, straightforward normal, and max score is higher than that of front-facing looking or left-looking sclera acknowledgment. Notwithstanding, we can pick either a quality-based normal or a straightforward normal on the off chance that we need a higher Veritable Acknowledge Rate [5] (GAR) when FAR diminishes. The maximum score combination technique can be utilized to bring down FAR when GAR arrives at 1.

8. The data to be tested for ScleraVO is gathered from 30 candidates on different days and times. More than 500 pictures of individuals are gathered every different day. The data chosen is filled with noise factors like reflection and luminosity to make the task a little bit challenging. The datasets ScleraVO and SBVPI are used to verify its verification and generalization.

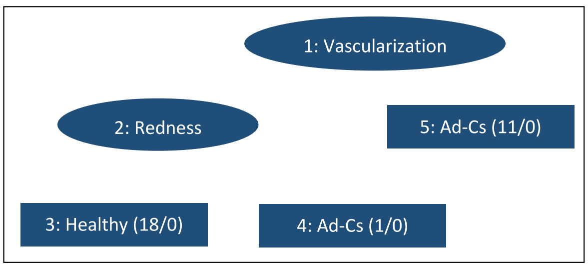


Fig. 7. Classification using Random Tree.

Obtained results using the ScleraVO dataset Fig. 10 shows some extracted ROI and vessels, and the vascular patterns are extracted. Because the mined blood vessels are distributed across various layers, their intensities vary. The subsequent recognition step compares the act of various algorithms using these results as inputs.

Various classification frameworks have been modified for comparisons among other sclera recognition systems. All the models were trained for the same approach to compare with consistency in results. Many networks that were studied as part of the proposed method are Inception Net, DialatedNet, NasNet, ScleraNet, and SLBNet. F1 score and accuracy were obtained in each of the above methods. Inception net gives an accuracy of 94.27 % with an accuracy of 93.27 %, DialateNet gives an accuracy of 94.27 % with an F1 score of 93.72 %, NasNet gives an accuracy of 80.25 % with F1 Score of 80.26 %, ScleraNet give an accuracy of 92.62 % with F1 Score of 92.61 % and the proposed method SLBNet gives an accuracy of 96.91 % and F1 Score of 96.91 %. The Time Complexity of each of the networks was tested using the available datasets. The time taken in milliseconds by InceptionNet is 48.2 ms for 6.8M number of parameters, DialatedNet takes 45.8 ms for 11.0M parameters, NasNet takes 53.2 ms for 2.2M parameters, ScleraNet takes 33.4 ms for 5.0M parameters and the proposed method SLBNet takes 33.4 ms for 2.1M parameters.

9. The proposed method was tested using 11165 eye pictures, and some pictures were taken from CAS-PAL databases. The images for testing with different skin colours, illuminations, and distinct orientations with and without specs were carried out. The 3 LBPH-based Landscapes LBPH-BIN, LBPHBIN-DIST, and LBPH-DIST [8] were compared. The three features are used to extract histograms from each block. Therefore, the lengths of the 3 feature sets that were extracted are 68853, 1167, and 68853, respectively.

Based on the 3 feature sets, Ada-Boost had been used for training three classifiers. The trained classifiers' ROC curves are contrasted. It demonstrates that the LBPHBIN- DIST feature performs better than the other two features. The LBPH-BIN-DIST feature-based classifier has the highest identification rate of 98.3321 percent when the false positive rate reaches zero. There are two layers in the LBPH-BIN-DIST feature-trained cascaded Ada-Boost classifier. While the first layer employs only seven features, it successfully throws out 96 % of negative training samples. From the initial 68853 features, 19 features are used in the final classifier. The presence of these chunks around the understudies, eye corners, and eyelids shows that these locales contain the most discriminative data for distinguishing eye positions. The subsequent eye recognition rate for an open eye was 99.5250 % and for shut eyes was 98.4772 %.

10. The proposed techniques utilize the Sclera Division and Acknowledgment Benchmarking Competition [12] dataset for exploratory testing. The pictures of the eye are taken with different cases like haze, flickering, and shut-eye. The images of the eye are taken from multiple angles like up, left, center, and right. The time complexity is linear for testing. Due to computational constraints, the model was run for 100 epochs, with a maximum accuracy of 87.65 percent, an average accuracy of 83.76 percent, and a minimum accuracy of 81.17 percent. The model's random initialization of weights is the cause of the variation in accuracy. The accuracy, recall, FRR, GAR, and FAR scores were once more compared to the model. It is been extremely fascinating to see the accuracy score of 0.86, for the sclera acknowledgment. The average scores for GAR, FRR, and FAR were 0.13 and 0.015, respectively. For sclera recognition, this demonstrates how deep learning models can be accomplished efficiently and have a lower error rate. Table 8 displays the comparison of the proposed model's results to those of other models presented by various authors in the SSRBC 2016 competition. Because very little work has been done to earn recognition, only two works are shown in the table. The participating team Sl. is one of the two works.

The primary framework in the table is a KNN-based framework for perceiving the sclera that can be executed in two simple tasks: highlight extraction and coordinating. To match the separated highlights to the preparation tuples by contrasting a specific test tuple with preparing tuples that are like it, they utilized the Histogram of Situated Slope (Hoard) descriptor for include extraction. Sl., the resistance group, No two utilizes a multiclass include-based classifier for acknowledgment. They utilized a 2D Gabor channel [12] that decreased the size of the component vector to 720 and a KNN for grouping, which prompted an increment of 1.56 % over the primary group. By training its network architecture, the proposed model makes use of CNN's knowledge of self-learning by convolving images at each layer and feeding the output to the next layers. By refreshing the loads through back engendering, the model figures out how to recognize and address its blunders. By correcting errors and producing better outcomes, this repetition aids the model in training effectively. Deep learning architectures make it possible to take this kind of approach. Time taken in testing is more complicated than training in traditional methods like KNN.

11. The experimental tests were performed on fifty eye images collected from distinct volunteers. The images were captured from four different angles up, down, left, and right so that the sclera-conjunctiva region could be gained. The results obtained through

(a) (b)

Fig. 8. Both front facing and side-looking picture matches are coordinated. The section matching outcomes for both Fig. (a) And (b) are 59 % and 70 %. The sclera designs are very much matched from 2 pictures of similar people.

Fig. 9. Front facing picture has bad quality and doesn't match well, yet the side-looking picture matches well overall. The section matching outcomes for both Fig. (a) and b) make up 10 % and 37 %, respectively. The sclera designs are all around paired in left-looking sclera pictures however not in front facing looking sclera pictures.

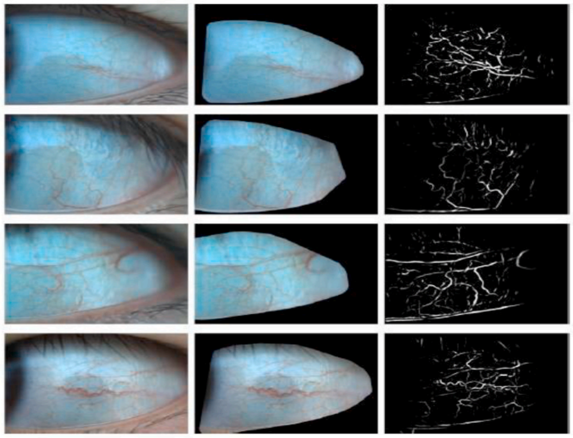


Fig. 10. A few consequences of the vessel division on the ScleraVO dataset. Left: Unique eye picture; Middle: return on initial capital investment region; Right: Sclera vasculature.

transforming from the original color image to a binary image by using methods such as grayscale processing, binary processing, noise removal, and automatic threshold choosing are shown below. The images are gathered by pre-processing from the original image of an eye, then segmenting the sclera part, and then it is converted to a grey scale image. Fig. 11 shows the conversion of the original image to a grayscale image.

The following Fig. 12 shows the resulting images after transformation. Image (A) can be seen that contains.

Fig. 12 contains the resulting images after transformation. In Image (A) can be seen that it contains a lot of noise. Image (B) is the result of applying Otsu's methods and isolated island removal without any noise. Image (C) is the result of edge detection [13] and digitization along with the edge parameter of blood vessels has been shown.

12. In Table 9, an examination is made among U-Net, wide U-Net, and U-Net++ regarding the number of boundaries and their division exactness across various errands, including lung knob division, colon polyp division, liver division, and cell cores division. The outcomes show that wide U-Net reliably beats U-Net as a rule, with the exemption being liver division, where the two designs perform in basically the same manner. This better exhibition of wide U-Net is credited to its bigger number of boundaries. U-Net++ without profound management shows huge presentation support contrasted with both U-Net and wide U-Net, bringing about a normal improvement of 2.8 and 3.3 places in Crossing Point over Association (IoU). Whenever profound management is applied to U-Net++, a typical improvement of 0.6 focuses is seen over U-Net++ without profound oversight. Eminently, profound oversight remarkably improves the precision of liver and lung knob division errands, however, it limitedly affects cell cores and colon polyp division. This disparity can be credited to the way that polyps and liver designs can differ in scale inside video casings and CT cuts, requiring a multi-scale approach including all division branches (profound oversight) to accomplish exact division. Fig. 2 gives a visual correlation of the outcomes obtained utilizing U-Net, wide U-Net, and U-Net++.

Table 8
Model comparison.

S No.	Propose model vs Existing models	Accuracy
1	Arun Kumar S V(SJCE, Mysuru, Karnataka, India)/segmentation and recognition (SSRBC 2016)	80.55
2	Chandranath Adak (Griffith University, Australia) & Bhagesh Seraogi (ISI, Kolkata, India)/recognition (SSRBC 2016)	82.11
3	Proposed Model	87.65

13. In this review, thorough situation testing was directed, enveloping different CNN boundaries and various preprocessing procedures. In the underlying situation, an examination of a few enhancer boundaries, to be specific Adam, Adam, SGD, and RMSprop, was embraced to decide the best analyzer. Each streamlining agent boundary was tried across the first dataset, expanded dataset, increased dataset exposed to grayscale preprocessing, and increased dataset exposed to thresholding preprocessing. A proper learning rate boundary of 0.0001 and a cluster size of 32 were utilized in this stage.

Consequently, the subsequent situation included a correlation of learning rate boundaries considering the still up in the air in the main situation, utilizing a group size of 32. The learning rate values analyzed included 0.01, 0.001, 0.0001, 0.00001, and 0.000001. Like the main situation, each enhancer boundary was tried across the first dataset, increased dataset, expanded dataset with grayscale preprocessing, and increased dataset with thresholding preprocessing.

The third situation zeroed in on looking at bunch size boundaries considering the ideal analyzer and gaining rate values acquired from the initial two situations. Bunch size upsides of 32, 64, and 128 were investigated, with each analyzer boundary being tried across the first dataset, increased dataset, expanded dataset with grayscale preprocessing, and expanded dataset with thresholding preprocessing.

In synopsis, the results of all situations were looked at, considering results from the first, expanded, and pre-processed datasets. This far-reaching approach is intended to recognize the EfficientNetB0 engineering CNN model with the best boundaries, including the best enhancer, clump size, and learning rate.

Table 10 shows the best boundaries of each dataset, specifically the best enhancer, the best learning rate, furthermore, the best group size. The best outcomes were acquired utilizing the grayscale dataset utilizing the Adam enhancer, learning rate 0.0001, clump size 32.

14. The outcomes from the analysis are displayed in Table 11. The grouping exactness acquired for the open and close eye order model, left, right, focus characterization model, and left, right, focus, and all-over arrangement model are displayed in Table 11. For the open and close eye grouping, the CNN model was prepared for 15 ages and exit 0.5 was finished to try not to over-fit as the dataset was not enormous. For the left, right, focus eye development grouping, the CNN model was prepared for 50 ages. The CNN model was trained for 200 epochs to classify the left, right, center, top, and down eye movements. The grouping exactness of the open and close eye characterization model which was utilized for flicker identification is 98 %. The approval precision for this model is close to 100 %. The grouping exactness of the 3-class (left, focus, right) classification model was higher than that of the 5-class (left, focus, right, up, down) characterization model. The classification of the three classes is 96 % accurate. The approval precision for this situation is 98 %. The exactness in 5-class characterization is 90 %. The approval accuracy for this situation is 89 %. The expectation scores from the two eyes are utilized to get the last result expectation. The arrangement without utilizing scenes was more exact than utilizing displays. The histogram evening out finished in the pictures assisted with expanding the exactness as it assisted with separating the iris and the sclera more, which assumes a significant part in the development order. The lighting conditions and the quality of the experiment's camera also affect the classification.

15. Table 12 shows the particular presentation measurements for Deeplabv3+, UNet++, Res-U2Net and Res-U2Net-Light, and it very well may be seen that the Res-U2Net metric works best among all the examination organizations. Fig. 10 depicts the three networks' superiority and intuitively demonstrates that the Res-U2Net network achieves the optimal eye segmentation index in comparison to the other networks.

16. To assess the exhibition of the calculation for sclera division, every one of the 218 images from the dataset was physically fragmented; these divisions were utilized as ground truth and contrasted and the veils acquired through the cycle. The precision, recall, accuracy, F1-score, and Jaccard metrics were used to evaluate performance, and the results are presented in Table 13.

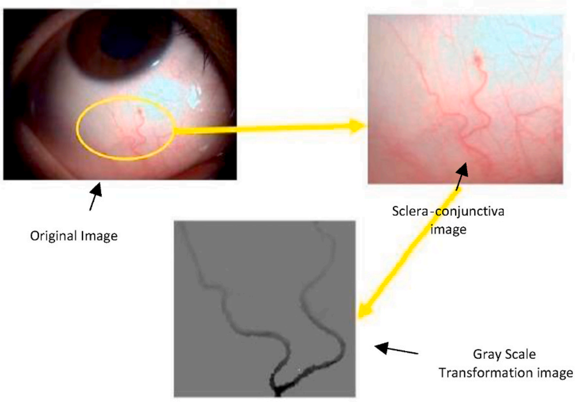


Fig. 11. Original to Grayscale image conversion.

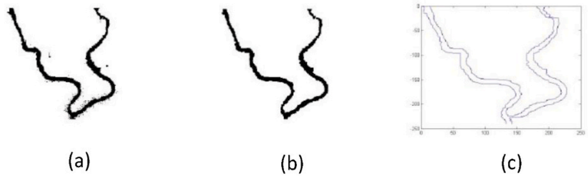


Fig. 12. Noisy image, the result of Otsu's methods, and image without noise.

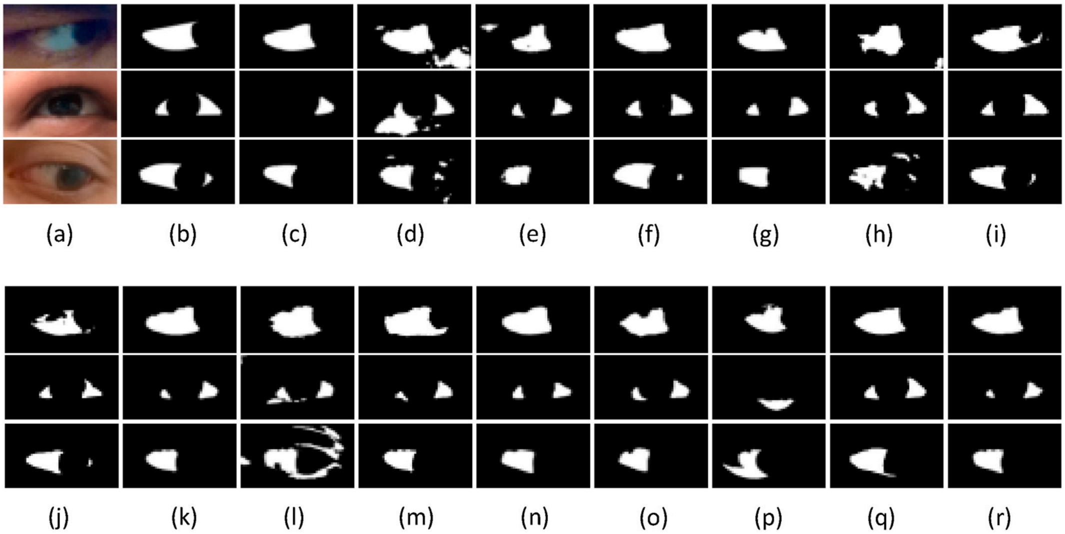


Fig. 13. A) Original image; (b) Ground truth mask; and the submitted binary masks from: (c) AB Sclera Net, (d) CGANs2020CL, (e) Color RITNet, (f) FCN8, (g) Mask2020CL, (h) MU-Net, (i) Multi-Deeplab, (j) Multi-FCN, (k) RGB–SS–Eye-MS, (l) SNet, (m) SaSSNet, (n) ScleraMaskRCNN, (o) ScleraU-Net, (p) SSIP, (q) UNet-P, (r) Y-SS-Eye-MS.

Table 9
Results of segmentation (IoU: %) for U-Net, wide U-Net, and our recommended engineering UNet++ with and without profound management.

Architecture	Params	Dataset			
		cell nuclei	Colon polyp	liver	Lung nodule
U-Net	7.76M	90.77	30.08	76.62	71.47
Wide U-Net	9.13M	90.92	30.14	76.58	73.38
U-Net++ w/o DS	9.04M	92.63	33.45	79.70	76.44
U-Net++ w/DS	9.04M	92.52	32.12	82.90	77.21

These outcomes can change contingent upon the tuning of the hyperparameters of the pipeline. Referring to the highest precision with the highest recall, which prevented the inclusion of zones from the palpebral conjunctiva, was how empirically found the best combination, as shown in Table 14. Truth be told, we will separate variety highlights from the sclera, which would be less impacted assuming something is forgotten about for any model because the sclera tone is as yet unchanged. As depicted segments with low recall typically lack some thin or shadowed areas. Nonetheless, even for this situation, like the instance of vessels, the whole sclera doesn't need to be portioned with the highest level of accuracy; all things considered, the divided part is uniform in variety, liberated from shadows or different clamors (cilia, and so forth.); subsequently, it is a decent delegate for the extraction of variety qualities.

17. Normal execution scores were determined over the division covers submitted to assess the submitted models. Certainty gauges for the detailed midpoints were likewise determined for all examinations by parceling the test information into 5 information parts (in a subject-disjoint way) and processing standard deviations over the five information parts. Results on masks for binary segmentation Double division veils are regularly created by division models through a thresholding methodology (which might be coordinated in the model) that commonly characterizes the compromise between n accuracy and review and decides a proper F1 score. Such double veils are normally the default result of contemporary (profound) division models. The main examination in this segment, subsequently, looks at the submitted models given scores from the double covers. UNet-P is the best entertainer

of the opposition however with a general outcome extremely near FCN8 considering the F1 and IoU scores. Five additional models bring about F1 scores above 0.8, i.e., RGB–SS–Eye-MS, Y-SS-Eye-MS, SaSSNet, Multi-DeepLab and CGANs2020CL. The following five arrangements, i.e., ScleraU-Net, Stomach muscle Sclera Net, Variety RITNet, ScleraMaskRCNN, and Multi-FCN, actually produce serious outcomes with F1 scores above 0.76, yet are fairly behind the top entertainers of SSBC 2020. Among the profound learning models, Mask2020CL accomplishes an F1 score of 0.717, while MU-Net and S-Net outcome in less aggressive execution pointers with F1 upsides of 0.651 and 0.462, individually. SSIP produces more vulnerable outcomes than most different models with an F1 score of 0.595. In any case, SSIP is the main SSBC 2020 methodology not given profound learning. It is intriguing to take note that the exhibition of the models isn't connected with their size/intricacy. UNetP, for instance, is among the more modest models with nearly 2 million boundaries, while FCN8 and Multi-FCN are the biggest models with around 135 million boundaries. The previous two of these models are the top entertainers of the opposition, while the last option is less aggressive. In outline, the outcomes show that most models delivered strong outcomes on the MOBIUS pictures, yet additionally, there is impressive changeability in the outcomes among the best and most obviously terrible performing models, regardless of whether comparative model geographies were considered for the division arrangements. To get a better knowledge of the presentation of the submitted models, Fig. 13 revolves around the submitted probabilistic division expectations.

3. Proposed work

3.1. Methodology

3.1.1. Training process

The data was split into normal images and corresponding mask images. The batch for the training segmentation model is 32 using monitor metrics of accuracy and loss to assess model performance. The hyperparameter-like learning rate was ket 0.1 for the training model, and the callbacks were used as an optimization technique to avoid overfitting (see Fig. 14).

3.1.2. Image loading and processing

The method begins with loading eye images with corresponding mask images. Following is the image processing.

- Image Resizing: Images are resized to a fixed size of 256x256 using OpenCV
- Normalization: Image data is normalized by dividing by 255.0 to scale pixel values between 0 and 1.
- Channels Handling: Channels-first ordering is enforced (backend.set_image_data_format('channels_last') commented out as channels-first ordering is expected).
- Data Loading: Images and masks are loaded from directories using os. listdir.

3.1.3. Image segmentation

Two segmentation models have been trained for the comparative analysis. The First segmentation model is prepared to utilize U-net, which includes an encoder-decoder structure with skip associations that interface relating layers between the two parts of the organization. As shown in Fig. 15 this design allows U-Net to effectively segment objects by preserving spatial information and is relatively straightforward to implement. U-Net is a well-established and widely used architecture that performs admirably in many segmentation tasks. However, it may struggle to capture fine details and handle objects with diverse scales.

The Second segmentation model is trained using the U-net++ architecture, which is an extension of U-Net that introduces a more intricate and nested skip connection architecture. As shown in Fig. 16 within the encoder and decoder sections, U-Net++ incorporates multiple skip pathways with varying degrees of nesting. This innovation allows the model to capture complex relationships and features at multiple scales, which can be particularly advantageous when dealing with objects of varying sizes and shapes. U-net may struggle to capture fine details and handle objects with diverse scales as effectively as U-Net++. U-Net++ is designed to excel in such scenarios, offering state-of-the-art results in complex segmentation challenges and providing improved accuracy when fine-grained details are essential.

Utilizes the unet_plus_2d model from keras_unet_collection. Binary cross-entropy loss is used for model compilation (model_Unet_plus.compile(loss = 'binary_crossentropy', optimizer = Adam(lr = 1e-3), metrics = ['accuracy', losses.dice_coef]) Accuracy and Dice coefficient are used as evaluation metrics. Adam optimizer with a learning rate of 0.001 is utilized (Adam(lr = 1e-3)). The model is trained using the model. fit with validation data and batch size of 8.

Table 10
Recreation consequences of best model for grayscale dataset.

Accuracy	79.22
Precision	80.3
Recall	79.22
F1-Score	78.87

Table 11
Results.

Classification type	Accuracy
Open and close eye	98 %
Left, right, center eye movements	96 %
Left, right, center, up and down eye movements	90 %

Table 12
Comparison with algorithms based on CNN.

Model	Recall (%)	Precision (%)	Miou (%)	MAE	F1-Score (%)
UNet++	95.6	92	95.5	0.066	92.80
Deeplabv3+	97.83	98.30	95.9	0.057	98.20
U2Net	99.05	98.48	97.30	0.0147	98.90
Res-U2Net	99.10	99.25	97.8	0.009	99.09
Res-U2Net-Lite	99.10	98.40	97.20	0.012	98.90

Table 13
Scleral segmentation algorithm performance scores.

	Precision	Recall	F1	Accuracy	Jaccard
Average	88.53	82.53	84.10	75.58	94.2
Std dev	13.29	12.10	9.39	12.58	4.25

Table 14
Top 3 performing algorithms in SSBC.

Segmentation Model	F1	Precision	Recall
UNet-P	86.8	90.9	83.1
FCN8	85.4	82.0	89.0
RGB–SS–Eye-MS	83.6	91.7	76.9

3.1.4. *Image concatenation*

The segmentation results from the image segmentation are used for image concatenation. Image concatenation in machine learning refers to the process of combining multiple images into a single image. This can be done along different axes, such as stacking images vertically or horizontally. It is an important step because it helps in achieving the region of interest of the image, in this study the region of interest is the sclera part of the eye. The resulting images after image concatenation are used for training the classification model (see Fig. 17).

3.1.5. *Image classification & prediction*

The classification model is trained using the CNN architecture. The concatenated images are being used for training the classification model (see Fig. 17).

3.2. *Environmental setup*

3.2.1. *Hardware and software*

No extra hardware extensions were utilized. The machine learning model was trained using Google Colab's free-tier instance, which utilized a TPU and 12 GB of RAM on the cloud. The hardware was utilized for debugging, performance evaluation, and testing

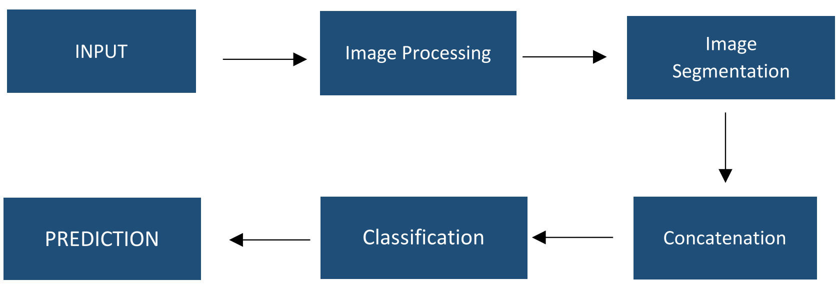


Fig. 14. Architecture of proposed work.

the behavior of the proposed system.

3.2.2. Dataset

The Dataset used in the training process is the SBVPI dataset which contains approximately 1600 images of the eye along with the masked images. The dataset contains images taken from different angles. The dataset is accessible via the SBVPI website and can be used for research purposes without any legal requirements. Some more images are collected from other datasets and non-copyright image websites to bring diversity to the training dataset. The images include normal eye images as well as conjunctivitis eye images.

This dataset with masked images and normal eye images is used for training the segmentation model, while the resulting images from the segmentation model are concatenated with the original images, and then the concatenated images are used for training the classification model.

3.2.3. Evaluation matrics

The following evaluation metrics were used to measure the performance of the machine learning model.

a. Average Training Loss

The average training loss metric represents the average loss of the machine learning model across the training dataset. The concept of loss refers to the discrepancy between the predictions made by a model and the actual values. A lower training loss suggests that the model is effectively performing on the training data. It is important to strike a balance between avoiding overfitting and achieving good performance on the validation dataset. The model's average loss is 0.0583. The formula for calculating the average training loss is given as equation (4.1).

$$Average\ Training\ Loss = \frac{1}{N}\sum_{i=1}^{N} Loss_{\mathrm{i}}$$

b. Validation accuracy

Validation accuracy is a metric that quantifies the percentage of accurate predictions out of the total number of cases analyzed. The high validation accuracy of 97.88 % indicates that the model demonstrates strong performance on the validation set, which is a distinct dataset that was not utilized during the training process. The formula for calculating the validation accuracy is given as equation 4.2

$$Validation\ Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ prediction}$$

c. Precision

Precision is determined by calculating the ratio of true positive predictions to the total positive predictions. This ratio is obtained by adding the number of true positives and false positives together. An accuracy rate of 98.28 % indicates that the model's positive class

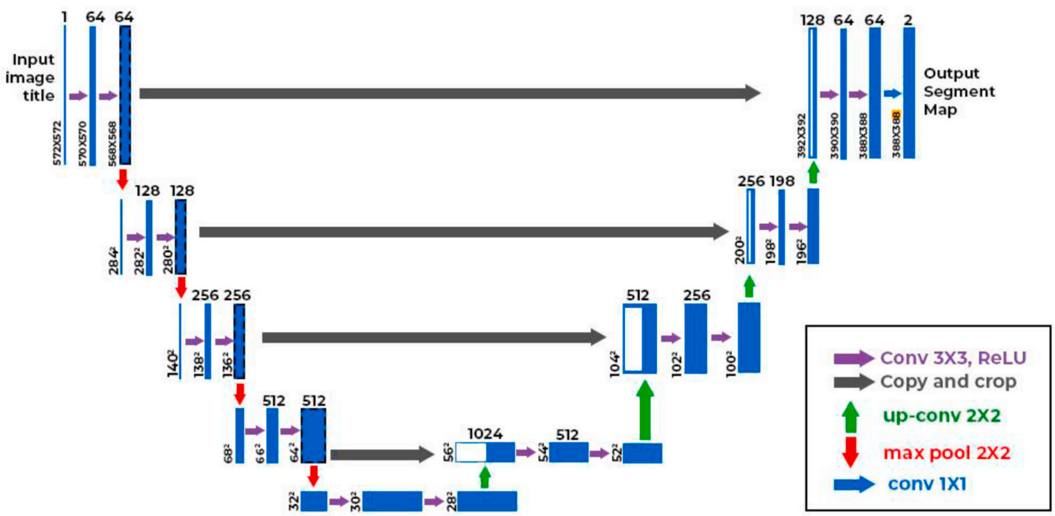


Fig. 15. U-net Architecture Image (Image taken from geeksforgeeks).

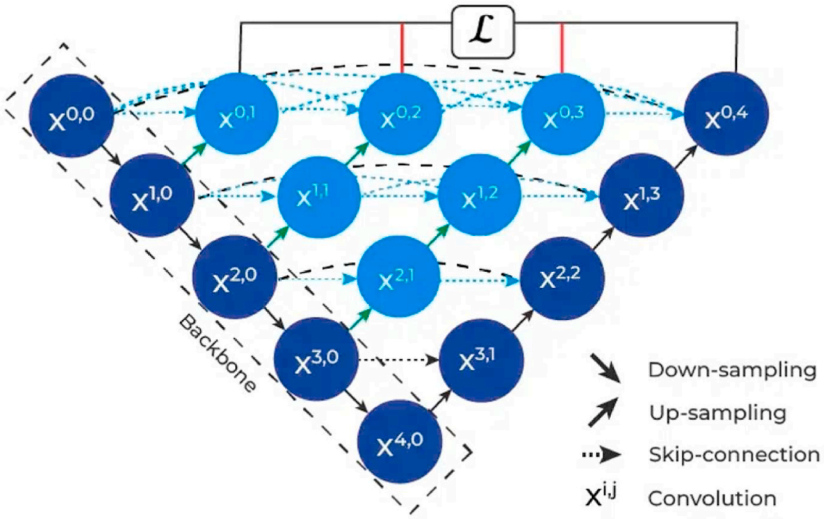
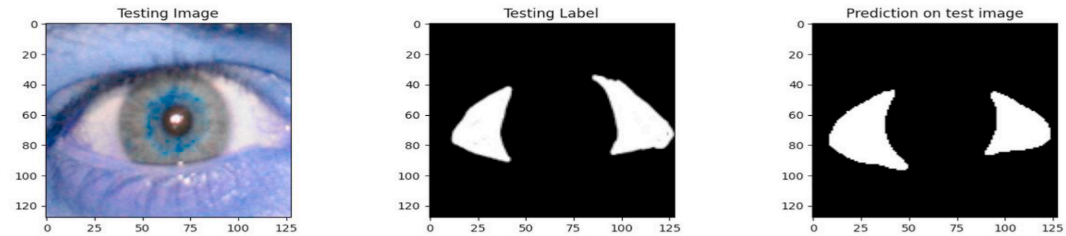
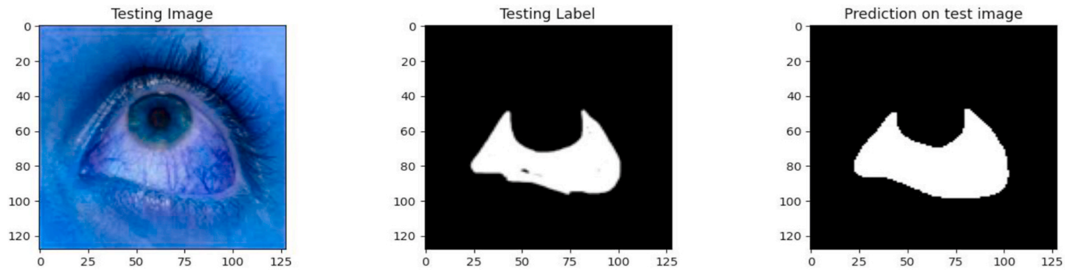


Fig. 16. Unet++ Architecture (Image taken from geeksforgeeks).



(a) U-net Segmentation results



(b) U-net++ Segmentation results

Fig. 17. (a) U-net Segmentation results. (b) U-net++ Segmentation results.

predictions are correct almost all of the time. This holds significant importance in situations where the consequences of false positives carry a substantial financial burden. The formula for calculating precision is:

$$Precision = \frac{True\ Positives}{True\ Posititve + False\ Positives}$$

d. Recall

Recall measures the proportion of actual positives that were correctly identified (true positives) out of all actual positives (the sum of true positives and false negatives). An accuracy rate of 97.45 % indicates that the model accurately identifies 97.45 % of all positive cases. Ensuring a high recall rate is of utmost importance in situations where the consequences of missing a positive result are

Table 15
Results of models trained using U-net architecture and U-net++ architecture.

Architecture	Accuracy	Loss	Validation Accuracy	Validation Loss	Dice Coefficient	Validation Dice coefficient
U-net	**0.9648**	**0.0195**	**0.9570**	**0.1172**	**0.9443**	**0.8472**
U-net++	**0.9707**	**0.0228**	**0.9401**	**0.2376**	**0.9364**	**0.7769**

significant, such as in disease diagnosis. The formula for the recall is:

$$Precision = \frac{True\ Positives}{True\ Posititve + False\ Negatives}$$

e. F1 Score

The F1-score represents the balanced combination of precision and recall, calculated as the harmonic mean. The metric combines precision and recall into a single number, offering a balanced assessment of the model's performance, particularly in cases of imbalanced class distribution. The F1 score of 97.86 % is indicative of a highly balanced model that demonstrates both high precision and recall.

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

4. Proposed work results

4.1. Image segmentation results

The image segmentation task is undertaken using the U-net and U-net++ architecture. The proposed work focused on performing and training a segmentation model in U-net++ architecture. Table 15 below shows the accuracy, loss, validation accuracy, validation loss, dice coefficient, and validation dice coefficient. The model achieved an accuracy of 97.48 %, showcasing its ability to accurately classify pixels in the segmented images. Furthermore, the low loss value of 0.0195 during training suggests that the model effectively minimized the error between the predicted and actual segmentation masks. When assessed on a separate validation dataset, the model exhibited a validation accuracy of 94.43 %, underlining its capability to generalize to unseen data. Additionally, the Dice coefficient, a critical metric for image segmentation tasks, yielded an impressive result of 94.43 % during training and 84.72 % on the validation set, further confirming the model's proficiency in accurately delineating objects of interest in images. The relatively low validation loss of 0.1172 underscores the model's robustness and proposes that it didn't overfit the preparation information.

The U-Net++ model achieved an accuracy of 97.07 %, reflecting its impressive capability to correctly classify pixels within the segmented images. The training loss, with a value of 0.0228, suggests that the model effectively minimized the error between predicted and actual segmentation masks. While the validation loss of 0.2376 on an independent dataset was slightly higher, it still points to the model's ability to generalize to previously unseen data, albeit with a minor drop in performance. Additionally, the Dice coefficient, a key metric in image segmentation tasks, yielded a strong score of 93.64 % during training and 77.64 % on the validation set. Although the validation Dice coefficient was slightly lower, these results underscore the efficacy of our U-Net++ segmentation model and its potential for various applications in image analysis, despite a relatively higher validation loss.

The suggested unet++ architecture is competitive with all the other architectures that we have mentioned in Table 16.

The calculated metrics prove the ability of propped work to be efficient and meet the complexity of image segmentation with high precision, recall, and f1 score.

In Fig. 17 (a), the segmentation using U-net architecture results in a corresponding label mask, and the original image is given. And Fig. 17 (b), shows the results of eye image segmentation using U-net++ architecture.

The resulting image after segmentation is processed to gain the region of interest. The zone of interest is the sclera part of the eye. To have a better understanding of the idea, we take a picture of a normal and conjunctivitis eye.

Table 16
Proposed work comparison with the already existing architectures.

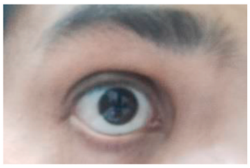
Architecture	Recall	F1	Precision
Unet-P [6]	94.2	94.3	94.4
Unet-2P [6]	93.9	94.0	94.1
tableUnet-4P [6]	93.8	93.9	94.0
RGB–SS–Eye-MS [16]	76.9	83.6	91.7
FCN8 [16]	89.0	85.4	82.0
Unet++(Proposed)	94.46	94.54	94.61

4.2. CNN results

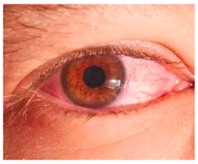
Convolutional Neural Networks algorithm is used for generating the predictions, the accuracy of diagnosis is 98.3 %.

4.3. Experiment result

The below tables show the results of experiments done on images gathered other than the dataset.
The below steps describe the experimental tasks using normal eye images.

Input normal Eye image	
Image segmentation result	
Image Concatenation	
Prediction	Normal Eye

The below steps describe the experimental tasks using the Conjunctivitis eye image.

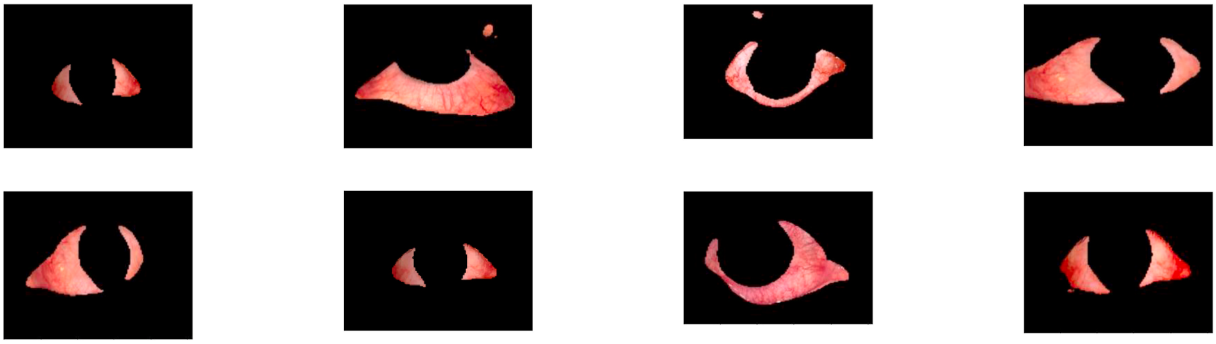
Input conjunctivitis image	
Image segmentation result	

(continued on next page)

(*continued*)

Image Concatenation	
Prediction	Conjunctivitis

More results on image segmentation.



5. Discussion and conclusions

The study of the proposed work showcases the Unet++ capability to perform image segmentation over the existing algorithms as shown in Table 10. The dataset was expanded by collecting more images from online resources. The CNN alone has better algorithms that can be utilized for diagnosis. In future work accuracy of prediction can be enhanced by using the latest techniques like embedding transformers, transfer learning, and deep learning.

Research data for this article

The dataset used in the research is open source and publicly available for research purposes. The dataset includes eye images, corresponding labeled masks, and the implementation code. The code can be accessed on our GitHub repository and for images SBVPI (other sources) dataset.

CRediT authorship contribution statement

E. Umamaheswari: Validation, Supervision, Project administration, Conceptualization. **Kanchana Devi V:** Validation, Supervision. **B. Sruthakeerthi:** Validation, Supervision, Conceptualization. **Nebojsa Bacanin:** Validation, Supervision. **Tushar Mathur:** Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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